Topics: 14.7 Maximum and Minimum Values; 14.8 Lagrange Multipliers
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1. For each function, find all critical points. Then use the Second Derivative Test to determine whether each critical point is a local maximum, local minimum, or saddle point.

   (a) $f(x, y) = 4x^2 + 9y^2 + 8x - 36y + 24$

   We first compute the first order partial derivatives:

   $$f_x(x, y) = 8x + 8, \qquad f_y(x, y) = 18y - 36.$$

   To find the critical points, we set both partial derivatives equal to zero:

   $$8x + 8 = 0, \qquad 18y - 36 = 0.$$

   Solving gives that the only critical point is $(-1, 2)$.
   Next we compute the second order partial derivatives:

   $$f_{xx}(x, y) = 8, \qquad f_{yy}(x, y) = 18, \qquad f_{xy}(x, y) = 0.$$

   Recall that

   $$D(x, y) = f_{xx}(x, y)f_{yy}(x, y) - \big(f_{xy}(x, y)\big)^2.$$

   At the critical point,

   $$D(-1, 2) = 8 \cdot 18 - 0^2 = 144 > 0.$$

   Since also

   $$f_{xx}(-1, 2) = 8 > 0,$$

   the point $(-1, 2)$ is a local minimum.

   (b) $g(x, y) = (y - 2)x^2 - y^2$

   We first compute the first order partial derivatives:

   $$g_x(x, y) = 2x(y - 2), \qquad g_y(x, y) = x^2 - 2y.$$

   To find the critical points, we set both partial derivatives equal to zero:

   $$2x(y - 2) = 0, \qquad x^2 - 2y = 0.$$

   From $2x(y - 2) = 0$, we have either $x = 0$ or $y = 2$.
   If $x = 0$, then the equation $x^2 - 2y = 0$ gives $y = 0$, so one critical point is $(0, 0)$.
   If $y = 2$, then the equation $x^2 - 2y = 0$ gives

   $$x^2 - 4 = 0,$$

   so $x = \pm 2$. This gives the critical points $(2, 2)$ and $(-2, 2)$.
   Therefore, the critical points are

   $$(0, 0), \qquad (2, 2), \qquad (-2, 2).$$

Next we compute the second order partial derivatives:

$$g_{xx}(x,y) = 2(y-2), \qquad g_{yy}(x,y) = -2, \qquad g_{xy}(x,y) = 2x.$$

At $(0,0)$, we have

$$D(0,0) = (-4)(-2) - 0^2 = 8 > 0.$$

Since also

$$g_{xx}(0,0) = -4 < 0,$$

the point $(0,0)$ is a local maximum.
At $(2,2)$, we have

$$D(2,2) = (0)(-2) - 4^2 = -16 < 0,$$

so $(2,2)$ is a saddle point.
At $(-2,2)$, we have

$$D(-2,2) = (0)(-2) - (-4)^2 = -16 < 0,$$

so $(-2,2)$ is a saddle point.

(c) $h(x,y) = (x+y^2)e^x$

We first compute the first order partial derivatives:

$$h_x(x,y) = e^x(x+y^2+1), \qquad h_y(x,y) = 2ye^x.$$

To find the critical points, we set both partial derivatives equal to zero:

$$e^x(x+y^2+1) = 0, \qquad 2ye^x = 0.$$

Since $e^x \neq 0$, these equations simplify to

$$x+y^2+1 = 0, \qquad y = 0.$$

Substituting $y = 0$ into the first equation gives

$$x+1 = 0,$$

so $x = -1$. Thus the only critical point is $(-1,0)$.
Next we compute the second order partial derivatives:

$$h_{xx}(x,y) = e^x(x+y^2+2), \qquad h_{yy}(x,y) = 2e^x, \qquad h_{xy}(x,y) = 2ye^x.$$

At the critical point,

$$D(-1,0) = \left(\frac{1}{e}\right)\left(\frac{2}{e}\right) - 0^2 = \frac{2}{e^2} > 0.$$

Since also

$$h_{xx}(-1,0) = \frac{1}{e} > 0,$$

the point $(-1,0)$ is a local minimum.

2. Consider the function $g(x,y) = x^3 - 3x^2 + 3x - y^2 + 4y$.
   (a) The function g has exactly one critical point. Find it!

We compute the first order partial derivatives:

$$g_x(x,y) = 3x^2 - 6x + 3 = 3(x-1)^2, \qquad g_y(x,y) = -2y + 4 = -2(y-2).$$

To find the critical points, we set both partial derivatives equal to zero:

$$3(x-1)^2 = 0, \qquad -2(y-2) = 0.$$

This gives

$$x = 1, \qquad y = 2.$$

Therefore, the only critical point is $(1,2)$.

(b) Show that the Second Derivative Test is inconclusive at this critical point.

We compute the second order partial derivatives:

$$g_{xx}(x,y) = 6x - 6, \qquad g_{xy}(x,y) = 0, \qquad g_{yy}(x,y) = -2.$$

Recall that

$$D(x,y) = g_{xx}(x,y)g_{yy}(x,y) - \big(g_{xy}(x,y)\big)^2.$$

Thus,

$$D(x,y) = (6x-6)(-2) - 0^2 = -12x + 12.$$

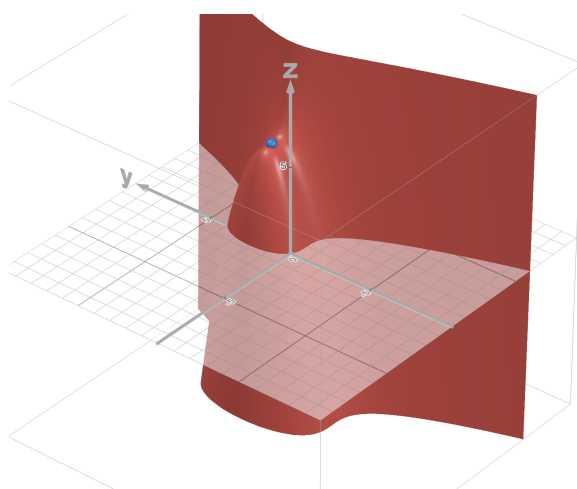
At the critical point $(1,2)$, we have

$$D(1,2) = -12(1) + 12 = 0.$$

Since $D(1,2) = 0$, the Second Derivative Test is inconclusive at $(1,2)$.

(c) Use a graphing utility (such as `Desmos 3D`) to plot the surface $z = g(x,y)$. Based on the graph, does this critical point appear to be a local maximum, local minimum, or saddle point?

The graph of $z = g(x,y)$ is shown below, with the point with x-, y- coordinates $(1,2)$ displayed in blue.



Based on the graph, the critical point appears to be a saddle point. Indeed, this can be verified

algebraically but that is not required here.

3. Find the absolute maximum and absolute minimum values of the function f on the set D.

(a) $f(x,y) = x^2 + y^2 + x^2y + 9$, where $D = \{(x,y) \in \mathbb{R}^2 : |x| \leq 1 \text{ and } |y| \leq 1\}$

Since D is closed and bounded, the absolute maximum and absolute minimum must occur either at a critical point in the interior of D or on the boundary of D.

We compute

$$f_x(x,y) = 2x + 2xy = 2x(1+y), \qquad f_y(x,y) = 2y + x^2.$$

Setting these equal to zero, we get

$$2x(1+y) = 0, \qquad 2y + x^2 = 0.$$

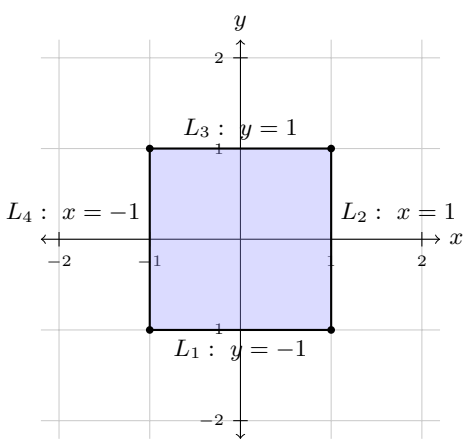
If $x = 0$, then $2y + 0 = 0$, so $y = 0$. If $y = -1$, then

$$2(-1) + x^2 = 0 \quad \Rightarrow \quad x^2 = 2,$$

which is not possible in D since $|x| \leq 1$. Thus the only critical point in D is $(0,0)$, and here we have

$$f(0,0) = 0^2 + 0^2 + 0^2(0) + 9 = 9.$$

Next we check the boundary.



- On L_1: $y = -1$, where $-1 \leq x \leq 1$, so

$$f(x,-1) = x^2 + (-1)^2 + x^2(-1) + 9 = 10.$$

Thus on L_1, both the minimum and maximum are

$$f(x,-1) = 10.$$

- On L_2: $x = 1$, where $-1 \leq y \leq 1$, so

$$f(1,y) = 1 + y^2 + y + 9 = y^2 + y + 10.$$

We compute

$$\frac{d}{dy}(y^2 + y + 10) = 2y + 1.$$

Setting this equal to zero gives

$$2y + 1 = 0 \quad \Rightarrow \quad y = -\frac{1}{2}.$$

Thus we check $y = -1$, $y = 1$, and $y = -\frac{1}{2}$:

$$f(1, -1) = 10, \qquad f(1, 1) = 12, \qquad f\left(1, -\frac{1}{2}\right) = \frac{39}{4}.$$

So on L_2, the minimum is

$$f\left(1, -\frac{1}{2}\right) = \frac{39}{4},$$

and the maximum is

$$f(1, 1) = 12.$$

- On L_3: $y = 1$, where $-1 \leq x \leq 1$, so

$$f(x, 1) = x^2 + 1 + x^2(1) + 9 = 2x^2 + 10.$$

We compute

$$\frac{d}{dx}(2x^2 + 10) = 4x.$$

Setting this equal to zero gives

$$4x = 0 \quad \Rightarrow \quad x = 0.$$

Thus we check $x = -1$, $x = 1$, and $x = 0$:

$$f(-1, 1) = 12, \qquad f(1, 1) = 12, \qquad f(0, 1) = 10.$$

So on L_3, the minimum is

$$f(0, 1) = 10,$$

and the maximum is

$$f(\pm 1, 1) = 12.$$

- On L_4: $x = -1$, where $-1 \leq y \leq 1$, so

$$f(-1, y) = 1 + y^2 + y + 9 = y^2 + y + 10.$$

We compute

$$\frac{d}{dy}(y^2 + y + 10) = 2y + 1.$$

Setting this equal to zero gives

$$2y + 1 = 0 \quad \Rightarrow \quad y = -\frac{1}{2}.$$

Thus we check $y = -1$, $y = 1$, and $y = -\frac{1}{2}$:

$$f(-1, -1) = 10, \qquad f(-1, 1) = 12, \qquad f\left(-1, -\frac{1}{2}\right) = \frac{39}{4}.$$

So on L_4, the minimum is

$$f\left(-1, -\frac{1}{2}\right) = \frac{39}{4},$$

and the maximum is

$$f(-1, 1) = 12.$$

Comparing all values, the absolute minimum is

$$f(0,0) = 9,$$

and the absolute maximum is

$$f(\pm 1, 1) = 12.$$

(b) $f(x,y) = x^2 + y^2 - 2x$, where D is the closed triangular region with vertices $(2,0)$, $(0,2)$, and $(0,-2)$.

Since D is closed and bounded, the absolute maximum and absolute minimum must occur either at a critical point in the interior of D or on the boundary of D.
We compute

$$f_x(x,y) = 2x - 2, \qquad f_y(x,y) = 2y.$$

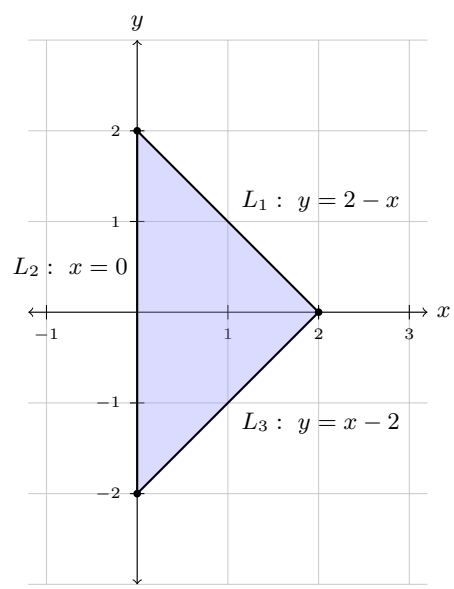
Setting these equal to zero, we get

$$2x - 2 = 0, \qquad 2y = 0.$$

Thus $x = 1$ and $y = 0$, so the only critical point in D is $(1,0)$, and here we have

$$f(1,0) = 1^2 + 0^2 - 2(1) = -1.$$

Next we check the boundary.



- On L_1: $y = 2 - x$, where $0 \le x \le 2$, so

  $$f(x, 2-x) = x^2 + (2-x)^2 - 2x = 2x^2 - 6x + 4.$$

  We compute

  $$\frac{d}{dx}\big(2x^2 - 6x + 4\big) = 4x - 6.$$

  Setting this equal to zero gives

  $$4x - 6 = 0 \quad \Rightarrow \quad x = \frac{3}{2}.$$

Thus we check $x = 0$, $x = 2$, and $x = \frac{3}{2}$:

$$f(0,2) = 4, \qquad f(2,0) = 0, \qquad f\left(\frac{3}{2}, \frac{1}{2}\right) = -\frac{1}{2}.$$

So on L_1, the minimum is

$$f\left(\frac{3}{2}, \frac{1}{2}\right) = -\frac{1}{2},$$

and the maximum is

$$f(0,2) = 4.$$

- On L_2: $x = 0$, where $-2 \leq y \leq 2$, so

$$f(0,y) = y^2.$$

This has minimum at $y = 0$, with

$$f(0,0) = 0,$$

and maximum at the endpoints $y = \pm 2$, with

$$f(0, \pm 2) = 4.$$

- On L_3: $y = x - 2$, where $0 \leq x \leq 2$, so

$$f(x, x-2) = x^2 + (x-2)^2 - 2x = 2x^2 - 6x + 4.$$

We compute

$$\frac{d}{dx}\big(2x^2 - 6x + 4\big) = 4x - 6.$$

Setting this equal to zero gives

$$4x - 6 = 0 \quad \Rightarrow \quad x = \frac{3}{2}.$$

Thus we check $x = 0$, $x = 2$, and $x = \frac{3}{2}$:

$$f(0,-2) = 4, \qquad f(2,0) = 0, \qquad f\left(\frac{3}{2}, -\frac{1}{2}\right) = -\frac{1}{2}.$$

So on L_3, the minimum is

$$f\left(\frac{3}{2}, -\frac{1}{2}\right) = -\frac{1}{2},$$

and the maximum is

$$f(0,-2) = 4.$$

Comparing all values, the absolute minimum is

$$f(1,0) = -1,$$

and the absolute maximum is

$$f(0, \pm 2) = 4.$$

4. A tennis ball manufacturer's monthly profit (in thousands of dollars) is modeled by

$$f(x, y) = 48x + 96y - x^2 - 2xy - 9y^2,$$

where x is the number of tennis balls manufactured (in thousands) and y is the number of hours of advertising purchased. The company can produce at most 50,000 tennis balls per month, and it can purchase at most 25 hours of advertising. Find the values of x and y that maximize profit, and find the maximum profit.

Let

$$D = \{(x, y) \in \mathbb{R}^2 : 0 \leq x \leq 50,\ 0 \leq y \leq 25\}.$$

Since D is closed and bounded, the absolute maximum of f must occur either at a critical point in the interior of D or on the boundary of D.

We compute

$$f_x(x, y) = 48 - 2x - 2y, \qquad f_y(x, y) = 96 - 2x - 18y.$$

Setting these equal to zero, we get

$$48 - 2x - 2y = 0, \qquad 96 - 2x - 18y = 0.$$

Equivalently,

$$x + y = 24, \qquad x + 9y = 48.$$

Subtracting gives

$$8y = 24 \quad \Rightarrow \quad y = 3,$$

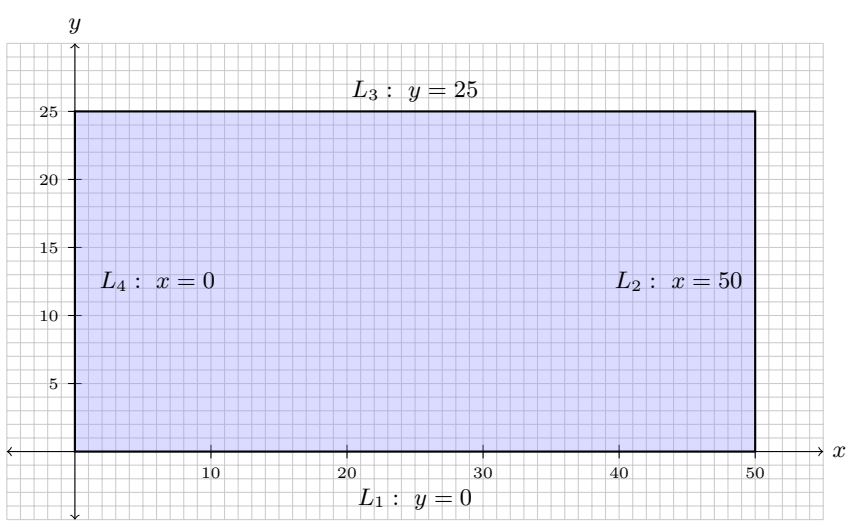
and then

$$x = 24 - 3 = 21.$$

Thus the only critical point in D is $(21, 3)$, and here we have

$$f(21, 3) = 48(21) + 96(3) - 21^2 - 2(21)(3) - 9(3)^2 = 648.$$

Next we check the boundary.



- On L_1: $y = 0$, where $0 \le x \le 50$, so

$$f(x, 0) = 48x - x^2.$$

We compute

$$\frac{d}{dx}(48x - x^2) = 48 - 2x.$$

Setting this equal to zero gives

$$48 - 2x = 0 \quad \Rightarrow \quad x = 24.$$

Thus we check $x = 0$, $x = 50$, and $x = 24$:

$$f(0, 0) = 0, \qquad f(50, 0) = -100, \qquad f(24, 0) = 576.$$

So on L_1, the maximum is

$$f(24, 0) = 576.$$

- On L_2: $x = 50$, where $0 \le y \le 25$, so

$$f(50, y) = 48(50) + 96y - 50^2 - 2(50)y - 9y^2 = -100 - 4y - 9y^2.$$

We compute

$$\frac{d}{dy}\big(-100 - 4y - 9y^2\big) = -4 - 18y.$$

Setting this equal to zero gives

$$-4 - 18y = 0 \quad \Rightarrow \quad y = -\frac{2}{9},$$

which is not in the interval $0 \le y \le 25$. Thus we check only the endpoints:

$$f(50, 0) = -100, \qquad f(50, 25) = -5825.$$

So on L_2, the maximum is

$$f(50, 0) = -100.$$

- On L_3: $y = 25$, where $0 \le x \le 50$, so

$$f(x, 25) = 48x + 96(25) - x^2 - 2x(25) - 9(25)^2 = -x^2 - 2x - 3225.$$

We compute

$$\frac{d}{dx}\big(-x^2 - 2x - 3225\big) = -2x - 2.$$

Setting this equal to zero gives

$$-2x - 2 = 0 \quad \Rightarrow \quad x = -1,$$

which is not in the interval $0 \le x \le 50$. Thus we check only the endpoints:

$$f(0, 25) = -3225, \qquad f(50, 25) = -5825.$$

So on L_3, the maximum is

$$f(0, 25) = -3225.$$

- On L_4: $x = 0$, where $0 \le y \le 25$, so

$$f(0, y) = 96y - 9y^2.$$

We compute

$$\frac{d}{dy}(96y - 9y^2) = 96 - 18y.$$

Setting this equal to zero gives

$$96 - 18y = 0 \quad \Rightarrow \quad y = \frac{16}{3}.$$

Thus we check $y = 0$, $y = 25$, and $y = \frac{16}{3}$:

$$f(0, 0) = 0, \qquad f(0, 25) = -3225, \qquad f\left(0, \frac{16}{3}\right) = 256.$$

So on L_4, the maximum is

$$f\left(0, \frac{16}{3}\right) = 256.$$

Comparing all values, the absolute maximum is

$$f(21, 3) = 648.$$

Therefore, the profit is maximized when the company manufactures 21 thousand tennis balls and purchases 3 hours of advertising. The maximum profit is \$648,000.

5. Use the method of Lagrange multipliers to find the maximum and minimum values of f subject to the given constraint.

(a) $f(x, y) = 9xe^y$, subject to $x^2 + y^2 = 2$.

Following the method of Lagrange multipliers, we solve

$$\nabla f = \lambda \nabla g \quad \text{subject to the constraint} \quad g(x, y) = 2,$$

where $g(x, y) = x^2 + y^2$. From $\nabla f = \lambda \nabla g$, we get

$$\begin{aligned} f_x &= \lambda g_x \qquad \Rightarrow \qquad 9e^y = 2\lambda x, \\ f_y &= \lambda g_y \qquad \Rightarrow \qquad 9xe^y = 2\lambda y. \end{aligned}$$

So the system is

$$9e^y = 2\lambda x, \qquad 9xe^y = 2\lambda y, \qquad x^2 + y^2 = 2.$$

Multiply the first equation by x to get

$$9xe^y = 2\lambda x^2.$$

Comparing with the second equation, we obtain

$$2\lambda x^2 = 2\lambda y.$$

Since $9e^y = 2\lambda x$ and $e^y \neq 0$, we have $\lambda \neq 0$. Hence

$$x^2 = y.$$

Substituting into the constraint,

$$x^2 + (x^2)^2 = 2 \quad \Rightarrow \quad x^4 + x^2 - 2 = 0 \quad \Rightarrow \quad (x^2+2)(x^2-1) = 0.$$

Thus $x = \pm 1$, and since $y = x^2$, we have $y = 1$.
So the candidate points are $(1,1)$ and $(-1,1)$. Evaluating:

$$f(1,1) = 9e, \qquad f(-1,1) = -9e.$$

Therefore, the maximum value is $9e$ at $(1,1)$, and the minimum value is $-9e$ at $(-1,1)$.

(b) $f(x,y) = 81x^2 + y^2$, subject to $4x^2 + y^2 = 9$

Following the method of Lagrange multipliers, we solve

$$\nabla f = \lambda \nabla g \quad \text{subject to the constraint} \quad g(x,y) = 9$$

where $g(x,y) = 4x^2 + y^2$. From $\nabla f = \lambda \nabla g$, we get

$$\begin{aligned} f_x &= \lambda g_x \qquad \Rightarrow \qquad 162x = 8\lambda x, \\ f_y &= \lambda g_y \qquad \Rightarrow \qquad 2y = 2\lambda y. \end{aligned}$$

So the system is

$$162x = 8\lambda x, \qquad 2y = 2\lambda y, \qquad 4x^2 + y^2 = 9.$$

Rearranging and factoring the second equation,

$$2y(1-\lambda) = 0,$$

so either $y = 0$ or $\lambda = 1$.
If $y = 0$, then the constraint gives

$$4x^2 = 9 \quad \Rightarrow \quad x = \pm\frac{3}{2}.$$

If $\lambda = 1$, then the first equation gives

$$162x = 8x \quad \Rightarrow \quad 154x = 0 \quad \Rightarrow \quad x = 0.$$

Substituting into the constraint,

$$y^2 = 9 \quad \Rightarrow \quad y = \pm 3.$$

Thus the candidate points are

$$\left(\frac{3}{2}, 0\right), \quad \left(-\frac{3}{2}, 0\right), \quad (0,3), \quad (0,-3).$$

Evaluating:

$$f\left(\frac{3}{2}, 0\right) = \frac{729}{4}, \qquad f\left(-\frac{3}{2}, 0\right) = \frac{729}{4},$$
$$f(0,3) = 9, \qquad f(0,-3) = 9.$$

Therefore, the maximum value is $\frac{729}{4}$ at $\left(\pm\frac{3}{2}, 0\right)$, and the minimum value is 9 at $(0, \pm 3)$.

6. Find the absolute maximum and absolute minimum values of the function

$$f(x, y) = 2x^2 - y^2 + 6y$$

on the disk $x^2 + y^2 \leq 16$.

Since $D = \{(x, y) : x^2 + y^2 \leq 16\}$ is closed and bounded, the absolute maximum and absolute minimum must occur either at a critical point in the interior of D or on the boundary of D.

We compute

$$f_x(x, y) = 4x, \qquad f_y(x, y) = -2y + 6.$$

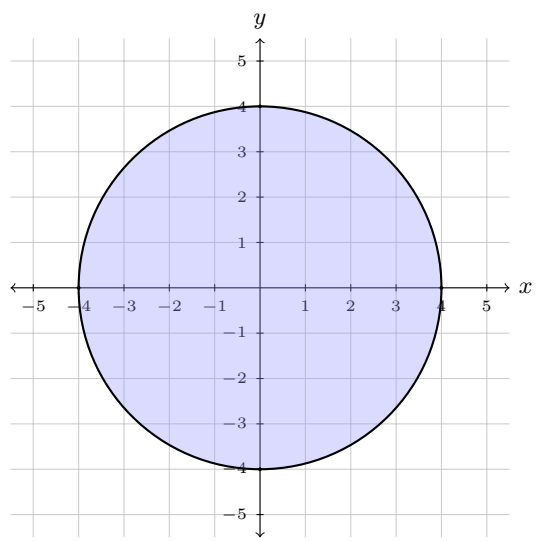
Setting these equal to zero gives

$$4x = 0, \qquad -2y + 6 = 0,$$

so the only critical point in D is $(0, 3)$. At this point,

$$f(0, 3) = 2(0)^2 - 3^2 + 6(3) = 9.$$

Next we check the boundary $x^2 + y^2 = 16$.



On the boundary, $x^2 = 16 - y^2$, so

$$f(x, y) = 2(16 - y^2) - y^2 + 6y = -3y^2 + 6y + 32.$$

Let

$$g(y) = -3y^2 + 6y + 32, \qquad -4 \leq y \leq 4.$$

Then

$$g'(y) = -6y + 6.$$

Setting this equal to zero gives

$$-6y + 6 = 0 \quad \Rightarrow \quad y = 1.$$

So we check $y = -4$, $y = 4$, and $y = 1$:

$$g(-4) = -40, \qquad g(4) = 8, \qquad g(1) = 35.$$

When $y = -4$, we have $x = 0$, giving the point $(0, -4)$.

When $y = 4$, we have $x = 0$, giving the point $(0, 4)$.

When $y = 1$, we have

$$x^2 = 16 - 1 = 15,$$

so $x = \pm\sqrt{15}$, giving the points $(\pm\sqrt{15}, 1)$.

Thus,

$$f(0, -4) = -40, \qquad f(0, 4) = 8, \qquad f(\pm\sqrt{15}, 1) = 35.$$

Comparing all values, the absolute maximum is

$$f(\pm\sqrt{15}, 1) = 35,$$

and the absolute minimum is

$$f(0, -4) = -40.$$

SOME USEFUL DEFINITIONS, THEOREMS, AND NOTATION

Critical Points and Local Extrema. Let f be a function of two variables, and let (a, b) be a point in the domain of f. We say that f has a *local maximum* at (a, b) if $f(a, b) \geq f(x, y)$ for all points (x, y) in the domain of f near (a, b). A *local minimum* is defined analogously. We say that (a, b) is a *critical point* of f if $f_x(a, b) = 0$ and $f_y(a, b) = 0$, or if at least one of these partial derivatives does not exist at (a, b). If f has a local maximum or local minimum at (a, b), then (a, b) is necessarily a critical point. A critical point that is neither a local maximum nor a local minimum is called a *saddle point*.

The Second Derivative Test. Suppose that the second order partial derivatives of f are continuous near a critical point (a, b). Define

$$D(a, b) = f_{xx}(a, b) f_{yy}(a, b) - \big(f_{xy}(a, b)\big)^2.$$

Then:

- If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$, then f has a local minimum at (a, b).
- If $D(a, b) > 0$ and $f_{xx}(a, b) < 0$, then f has a local maximum at (a, b).
- If $D(a, b) < 0$, then f has a saddle point at (a, b).
- If $D(a, b) = 0$, the test is inconclusive.

Absolute Extrema on Closed and Bounded Regions. If f is continuous on a closed and bounded region D, then f attains an absolute maximum value and an absolute minimum value on D. To find these absolute extrema, evaluate f at all critical points in the interior of D, and at all candidates for extrema on the boundary of D. The largest of these values is the absolute maximum value of f on D, and the smallest is the absolute minimum value.

Method of Lagrange Multipliers. To find the maximum and minimum values of $f(x, y)$ subject to the constraint $g(x, y) = k$ (assuming that such extrema exist, that f and g are differentiable, and that $\nabla g \neq \mathbf{0}$ on the constraint curve), we find all values x, y, λ such that

$$\nabla f(x, y) = \lambda \nabla g(x, y) \qquad \text{and} \qquad g(x, y) = k.$$

Then evaluate f at each corresponding point (x, y). The largest of these values is the maximum value, and the smallest is the minimum value.

Suggested Textbook Problems
Section 14.7: 1-56
Section 14.8: 1-29